CampusNotes - Academic Study Resource

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING - SEMESTER 5

OFFICIAL ACADEMIC STUDY MATERIAL - CSE CURRICULUM

Assignment 6: String Matching & NP-Completeness Reductions

Course: Design Analysis and Algorithms [23CS3502]
Category: Assignments | Assignment #6 | Academic Year: 2026-27

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1. Overview & Objective:

Pattern matching automatons, polynomial verification, and NP-hard problem reductions.

2. Assignment Problems & Instructions:

1. Compute the prefix function pi table for KMP string matching algorithm.
2. Explain Rabin-Karp algorithm rolling hash collision handling and complexity.
3. Define classes P, NP, NP-Complete, and NP-Hard with Venn diagram illustration.
4. Prove polynomial-time reduction from 3-SAT to Vertex Cover problem.

Submission Guidelines & Verification:

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